# Are short primality proofs hard to find?

## Elliptic curve certificates, from very short to one-shot

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Based on the survey *Elliptic curve primality proofs, from very short to one-shot* (written with Claude, Anthropic) and three open challenge repositories.

# Someone hands you a prime

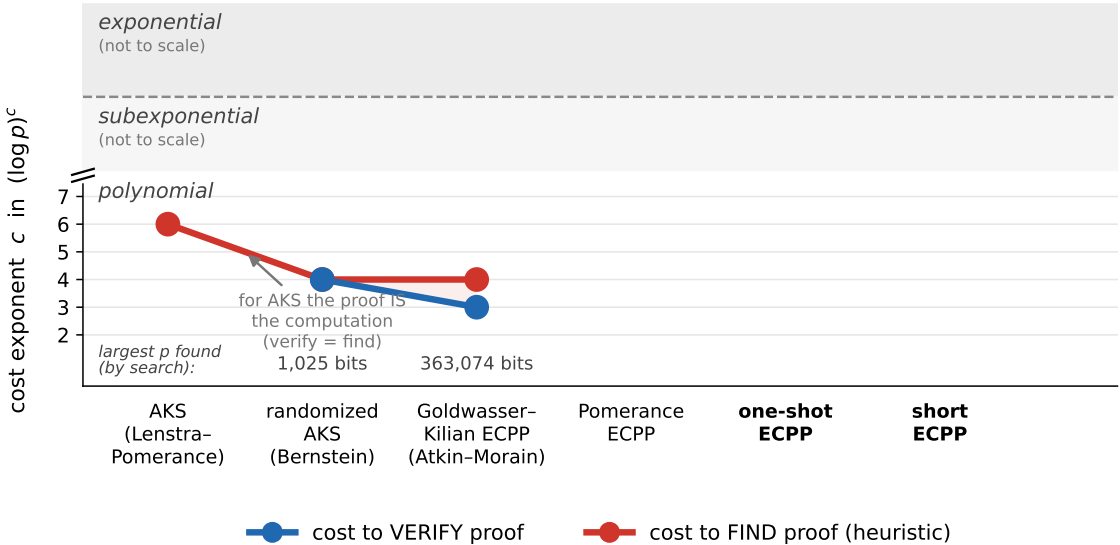
- Field primes of standard curves (P-256, Curve25519, secp256k1, BLS12-381), Diffie–Hellman groups, parameters chosen by someone else (possibly an adversary).
- Probabilistic tests prove nothing, and adversarially chosen composites have fooled real libraries.

  Albrecht, Massimo, Paterson, Somorovsky, *Prime and Prejudice*, CCS 2018.

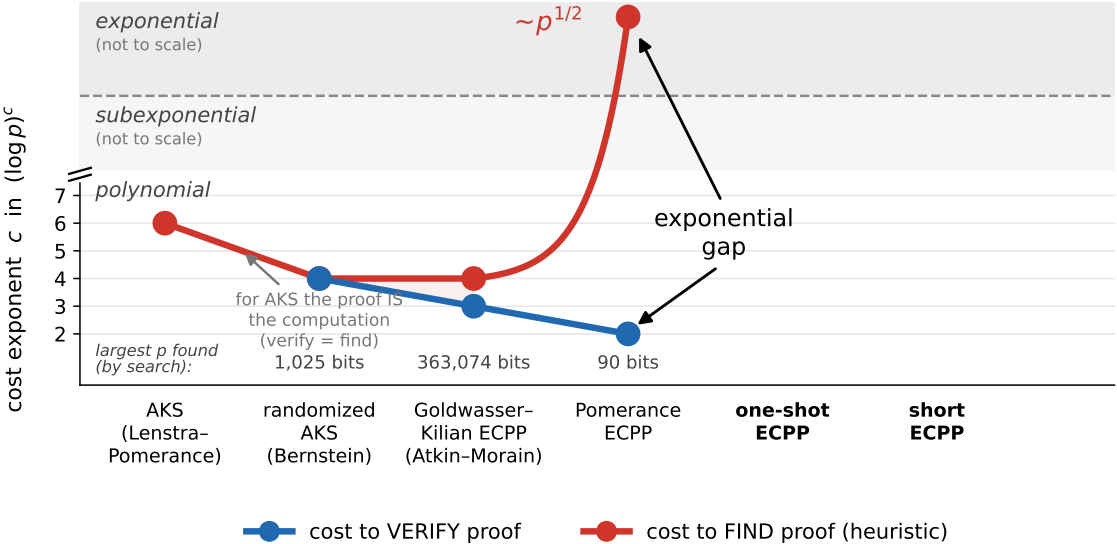
  OpenSSL 3 now runs $\geq 64$ Miller–Rabin rounds ($\geq 128$ above 2048 bits).
- A primality certificate for $p$: data whose validity is checked by an efficient *deterministic* algorithm and implies, *unconditionally*, that $p$ is prime.
- Two costs, as for any proof system:
  - FIND the certificate: once, by anyone; heuristic algorithms are fine;
  - VERIFY it: by everyone, every time; must be rigorous.

How cheap can verification be, and what does finding then cost?

# Find vs. verify



# Find vs. verify



# One lemma behind every elliptic certificate

**Lemma (Goldwasser–Kilian 1986, Pomerance 1987)**

Let $p > 3$ be odd and $L(p) = q + 1 + \lfloor 2\sqrt{q} \rfloor$ with $q = \lfloor \sqrt{p} \rfloor$. If $x_0$ is the $x$-coordinate of a point of order *exactly* $m > L(p)$ on $By^2 = x^3 + Ax^2 + x$ modulo every prime $\ell \mid p$, then $p$ is prime.

*Proof.* Otherwise some prime $\ell \le \sqrt{p}$ divides $p$, and $m \le \#E(\mathbb{F}_\ell) \le \ell + 1 + 2\sqrt{\ell} \le L(p) < m$. □

The verifier runs the Montgomery $x$-only ladder (as in X25519) modulo $p$:

- $[m](x_0 : 1) = (X : 0)$ with $\gcd(X, p) = 1$;
- $[m/q](x_0 : 1) = (X : Z)$ with $\gcd(Z, p) = 1$ for each prime $q \mid m$.

No need to check that $x_0$ is on $E$: it is on $E$ or its twist, and the ladder serves both.

The verifier must know the factorization of $m$. Each format below is a rule for how $m$ may be built so that it can.

# Attacking the verifier

- **The order of $P$ in $E(\mathbb{Z}/p\mathbb{Z})$ is not enough.**
  $p_0 = 4410667997551 = 2098153 \cdot 2102167$ has a point of order exactly $8 \cdot 525029 > L(p_0)$ in $E(\mathbb{Z}/p_0\mathbb{Z})$: order 8 modulo one factor and 525029 modulo the other.
  The gcd guards force order $m$ modulo *every* $\ell \mid p$; a failed guard factors $p$ (ECM, run in reverse).

- **The fold attack.** One-shot certificates list the primes $q_i \in (n^2, n^4)$ of $m$. Check them against $m$ rather than its $n^2$-rough part, and a repeated small prime $c$ can be folded into a fake "prime" $cq$.

- **Cost attacks.** The verifier's cost bound must hold over *all* certificates, including malicious ones. The August 2026 revision of short ECPP caps the smoothness bound and the radical of each filler: $O(n^2 \log n)$ bit operations, worst case, with no precomputation.

# Pomerance triples $(p, A, x_0)$: 2.5 Miller–Rabin rounds

Let $k$ be least with $2^k > L(p)$. Double $(x_0 : 1)$ $k$ times modulo $p$: $\gcd(Z_{k-1}, p) = 1$ and $Z_k \equiv 0$.

- Verify: $5k \approx \frac{5}{2} \log_2 p$ multiplications mod $p$, about 2.5 Miller–Rabin rounds.
- Exists for every prime $p > 3$ (Pomerance 1987; descend the 2-isogeny volcano to make the 2-Sylow subgroup cyclic).
- Find: $\#E$ must be one of the 2–4 multiples of $2^k$ in the Hasse interval, so $\widetilde{\Theta}(\sqrt{p})$ curves. Neither point counting nor CM helps. Best test: one ladder projects $P$ into the 2-Sylow subgroup, then double with early abort.

**DANGER3 challenge** (BIRS, April 2026); every entry a human + AI collaboration:

| $p$ | candidates | resources |
|---|---|---|
| $10^{19} + 51$ | $3.7 \times 10^9$ | 400 s, multithreaded C (Apr 2026) |
| $10^{24} + 7$ | $1.8 \times 10^{10}$ | 6 h, laptop (Jun 2026) |
| $10^{26} + 67$ | $1.4 \times 10^{11}$ | 45 min, one RTX 6000 GPU (Jun 2026) |
| $10^{27} + 103$ | $5.6 \times 10^{12}$ | 52 min, 16 RTX 5090 GPUs (Aug 2026) |

# One-shot ECPP: relax $2^k$ to "smooth"

**Certificate** $(p, A, x_0, m, q_1, \ldots, q_k)$ with $n = \lceil \log_2 p \rceil$: $m$ is $n^4$-smooth, $L < m < rL$ ($r$ the least prime of $m$), and $q_1 < \cdots < q_k$ are the primes of $m$ in $(n^2, n^4)$.

- Verify: trial-divide $m$ by the primes below $n^2$ (remainder trees); what is left must be exactly the $q_i$. No primality tests needed: a composite in $(n^2, n^4)$ has no divisors $> n^2$. $O(n^2(\log n)^2)$ bit operations.
- Find: need the $n^4$-smooth part of $\#E$ to exceed $L \approx \sqrt{p}$. By Dickman, this happens with probability

$$\rho(u/2)^{1+o(1)} = p^{-1/8+o(1)}, \qquad u = \frac{\ln p}{\ln n^4},$$

  so $p^{1/8+o(1)}$ with random curves (SEA) and $p^{1/4+o(1)}$ with the CM method. The $o(1)$ is very negative: $1/\rho(u/2) \approx 200$ at 256 bits.
- *Restricted:* no list, $m$ must be $n^2$-smooth. The exponent doubles to $1/4$: $\approx 3 \times 10^7$ candidates at 256 bits. A few hundred bits of listed primes buy a factor of $\sim 10^5$.

# Your favorite primes, certified

**Curve25519:** $p = 2^{255} - 19$, $n = 255$, $(n^2, n^4) = (65025,\ 4.2 \times 10^9)$

```
A  = 437239380964693581744653678253266551762067077077329106579272222259744945941308
x0 = 213115656186445074078169027384953638449830067523086529248868815733501378176 62
m  = 149 * 727 * 679409 * 770503 * 12519979 * 15032243 * 58136053    (129 bits)
```

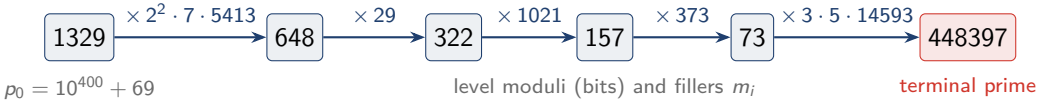
Listed: the five primes after 727. Verified in 5 ms (pure Python).

| prime | how |
|---|---|
| Curve25519, P-256, secp256k1 | CM search, 4–11 s on 16 cores |
| BLS12-381 | CM search, 18 s on 256 cores |
| P-384 | $p \equiv 3 \bmod 4$, so $y^2 = x^3 + x$ has $p + 1$ points; $p + 1$ has a 197-bit $n^4$-smooth part |
| P-521 | $p + 1 = 2^{521}$, so $y^2 = x^3 + x$ gives a Pomerance triple with $m = 2^{261}$ |
| $10^{120} + 79$ (generic, 399 bits) | CM search, minutes on 256 cores |

# Short ECPP: add recursion

**Certificate** $(p_0;\ A_0, x_0, o_0;\ \ldots;\ A_k, x_k, o_k)$; $n = \lceil \log_2 p_0 \rceil$ fixed, $B = \lceil n^2 / \log_2 n \rceil$.

- $o_i = m_i \cdot p_{i+1}$: $m_i$ is $B$-smooth with $\log_2 \mathrm{rad}(m_i) \le \lceil n / \log_2 n \rceil$; $p_{i+1}$ is $B$-rough with $B^2 < p_{i+1} < \sqrt{p_i}$.
- $L(p_i) < o_i < r_i L(p_i)$; point of order exactly $o_i$ modulo every prime $\ell \mid p_i$.
- Primality of $p_{i+1}$ is the next level's job. Stop at $p_{k+1} < B^2$: $B$-rough and below $B^2$ means 1 or prime.




Bits at least halve: $O(\log n)$ levels, $\lesssim 5n$ bits (6,328 here). Verify: $O(n^2 \log n)$ worst case, the order of a Pomerance triple. Find: subexponential.

# Why finding is subexponential

One level: find $E/\mathbb{F}_p$ with $\#E = S \cdot q$, $q$ prime $\approx\sqrt{p}$ (the next level).
The prover has to *see* $q$: the rest of $\#E$ must peel off by ECM within a budget $y$.

- candidates needed: $1/\rho(u)$, $u = \ln\sqrt{p}\,/\ln y$ (Dickman)
- cost per candidate: $L_y(1/2, \sqrt{2})$ (ECM)
- balance with $\ln y = \gamma(\ln p)^{2/3}(\ln\ln p)^{1/3}$; the total is $L_p\big(1/3,\ \frac{1}{6\gamma} + 2\sqrt{\gamma/3}\big)$, minimized at $\gamma = 12^{-1/3}$:

$$L_p\big(1/3,\ (3/2)^{1/3} \approx 1.14\big),$$

cheaper than factoring $p$ by the number field sieve, $L_p\big(1/3, (64/9)^{1/3} \approx 1.92\big)$, which is also what a Pratt certificate costs.

- The same constant for both supplies of orders: random curves + SEA, or CM. The CM discriminant scan costs $L_p(1/3, 0.76)$ and never dominates.
- Each level down shrinks the exponent by $2^{-1/3}$, so the top level dominates.

Heuristic: random-order model, Dickman, ECM. The $o(1)$ converges like $\ln\ln\ln p/\ln\ln p$, so constants are only indicative at feasible sizes.

# Records, and the cost of checking them

**Short ECPP records:**
$\mathrm{nextprime}(10^c)$ for
$c = 10, 20, \ldots, 400$.

- Largest: $10^{400} + 69$, 1329 bits, 5 levels, a 6,328-bit certificate.
- $c = 320, \ldots, 400$: $\approx$28,000 core-hours, $\approx$\$420 of AWS spot (192-core instances, six preemptions).
- For comparison: one-shot reaches 399 bits; Pomerance triples reach 90 bits.

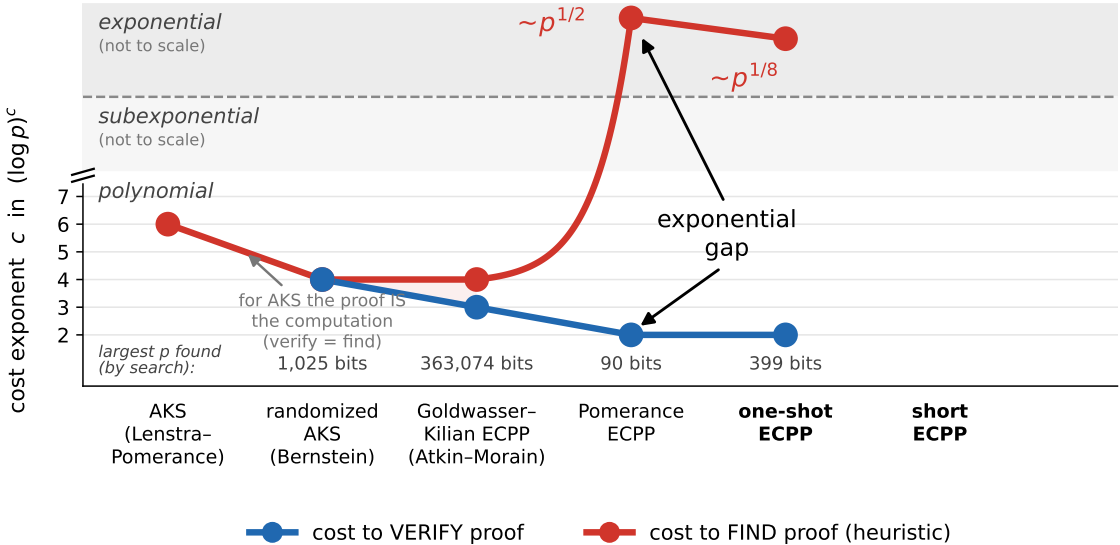
**Cost to verify** (Python; same big-integer arithmetic as `pow`):

| certificate | bits | time | MR rounds |
|---|---|---|---|
| Pomerance triple | any | | 2.5[a] |
| one-shot, Curve25519 | 255 | 4.9 ms | 85[b] |
| short, $10^{200} + 357$ | 665 | 16 ms | 21 |
| short, $10^{310} + 733$ | 1030 | 47 ms | 27 |
| short, $10^{400} + 69$ | 1329 | 90 ms | 27 |

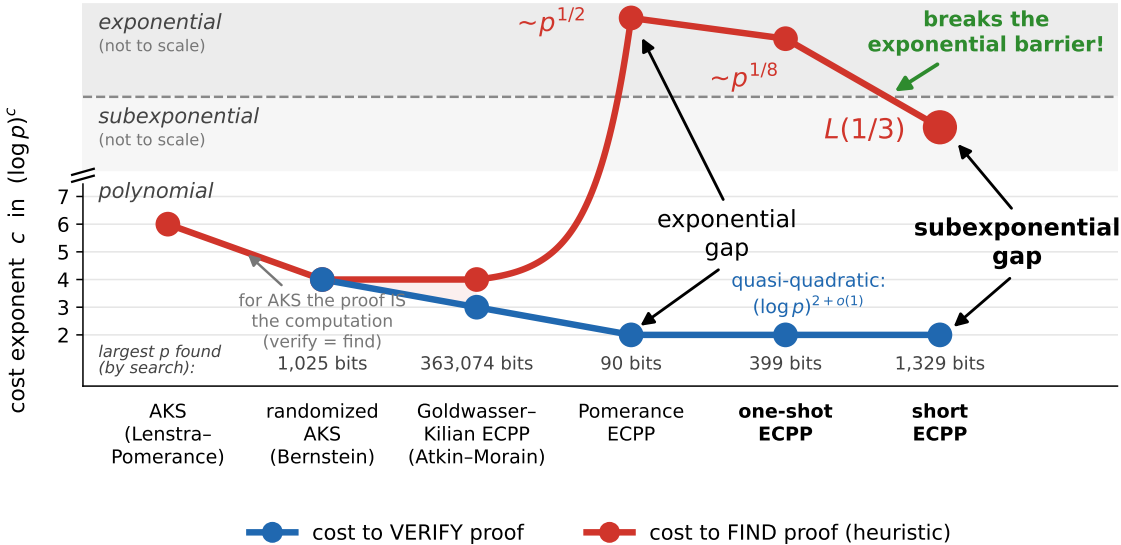
[a] exact multiplication count [b] dominated by sieving primes below $n^2$

An unconditional proof costs less than OpenSSL's 64 Miller–Rabin rounds.

# Are short proofs hard to find?



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# Open problems

1. Beat $\widetilde{\Theta}(\sqrt{p})$ for Pomerance triples, or explain why CM, isogenies, torsion families and sieving cannot: "$2^k \mid \#E$" has no multiplicative structure to sieve on.

2. One-shot via CM: close $p^{1/4}$ vs. $p^{1/8}$. This needs an output-sensitive enumeration of $4p = t^2 + |D|v^2$ *and* a root of $H_D$ mod $p$ in $h(D)^{1+o(1)}$ time.

3. The find/verify trade-off. Is there a certificate with (even heuristic) polynomial-time finding and $n^{2+o(1)}$ verification?
   - AKS–Bernstein: $(\log p)^{2+o(1)}$ to find, $(\log p)^{4+o(1)}$ to verify.
   - Short ECPP: $(\log p)^{2+o(1)}$ to verify, $L(1/3)$ to find.

4. Rigor: prove that a positive proportion of curves $E/\mathbb{F}_p$ have a $y$-smooth part of $\#E$ exceeding $\sqrt{p}$. Every finding bound here is heuristic.

5. Is $\frac{5}{2}\log_2 p$ multiplications optimal for general $p$? Pomerance declined to conjecture it.

# Thank you

- Survey: `github.com/AndrewVSutherland2/OneShotFastECPP`
  `reports/ecpp-varieties/ecpp-varieties.pdf`

- Challenges (definitions, reference verifiers, leaderboards):
  - `github.com/AndrewVSutherland/DANGER3`
  - `github.com/AndrewVSutherland/OneShotPrimalityProofs`
  - `github.com/AndrewVSutherland/ShortPrimalityProofs`

- Certificates welcome.

# Backup: the certificate families

| certificate | size (bits) | verify (bit ops) | find (heuristic) | exists for |
|---|---|---|---|---|
| Pratt | $O(n^2)$ | $O(n^3 \log n)$ | $L_p(1/3, 1.923)$ (NFS) | all $p$ |
| AKS–Bernstein | $n^{1+o(1)}$ | $(\log p)^{4+o(1)}$ | $(\log p)^{2+o(1)}$ randomized | all $p$ |
| Pomerance triple | $\approx 2n$ | $O(n^2 \log n)$ | $\widetilde{\Theta}(\sqrt{p})$ | all $p > 3$ |
| restricted one-shot | $\approx 2.5n$ | $O(n^2 (\log n)^2)$ | $p^{1/4+o(1)}$ SEA; $p^{1/2+o(1)}$ CM | all $p > 3$ |
| one-shot ($n^4$-smooth) | $\approx 2.5n$–$3n$ | $O(n^2 (\log n)^2)$ | $p^{1/8+o(1)}$ SEA; $p^{1/4+o(1)}$ CM | all $p > 3$ |
| short ECPP | $\lesssim 5n$ | $O(n^2 \log n)$ | $L_p(1/3, 1.14)$, SEA & CM | all $p \geq 5$ |
| classical ECPP | $\Theta(n^2/\log n)$ | $O(n^3)$ | $(\log p)^{6+o(1)}$ SEA; $(\log p)^{4+\epsilon}$ CM | all $p$ (heur.) |

Existence for every $p$ passes from Pomerance's theorem to every format containing his triples.

# Backup: two engines for candidate orders

- **SEA:** pick a random curve and count its points. Polynomial cost per candidate, flat forever (stock PARI: 7–14 core-seconds at 384–448 bits); both twists come free.
- **CM:** pick $D$ with $4p = t^2 + |D|v^2$; orders $p + 1 \pm t$ known before any curve exists. $\sim 10^7$ discriminants/s, but $|D| \le B$ yields only $\approx 0.45\sqrt{B}$ usable orders, so $K$ candidates cost $K^2$; the winner needs a class polynomial of degree $\approx \sqrt{|D|}$ (largest so far: $h = 35{,}085$).

**One-shot race**, same primes, 192-core machines:

| bits | $1/\rho(u/2)$ | CM wall | SEA wall | core-s/candidate (SEA vs CM) |
|---|---|---|---|---|
| 256 | $2.0 \times 10^2$ | 4.1 s | 68 s | — |
| 384 | $1.6 \times 10^4$ | 2,306 s | 615 s | 7.2 vs 3.5 |
| 416 | $4.8 \times 10^4$ | 1,774 s | 2,805 s | 10.1 vs 2.2 |
| 448 | $1.6 \times 10^5$ | 7,182 s | stopped at 20,035 s | 13.7 vs 4.5 |

Crossover extrapolates to $\approx$ 480–512 bits for one-shot and $\approx$ 2000 bits for short ECPP.

# Backup: restricted one-shot ($n^2$-smooth $m$, no list)

| bits | $u/2$ | $1/\rho(u/2)$ | SEA cost |
|---|---|---|---|
| 67 ($10^{20}$) | 2.74 | 12 | seconds, one core |
| 133 ($10^{40}$) | 4.71 | $1.3 \times 10^3$ | minutes, one core |
| 167 ($10^{50}$) | 5.62 | $1.7 \times 10^4$ | $\approx$1–2 core-hours |
| 200 ($10^{60}$) | 6.52 | $2.5 \times 10^5$ | $\approx 10^2$–$10^3$ core-hours |
| 256 | 8.00 | $3.1 \times 10^7$ | $\approx$5–10 core-years |
| 384 | 11.18 | $\approx 3 \times 10^{12}$ | out of reach |

First generic certificates: $10^{20} + 39$, $10^{30} + 57$, $10^{40} + 121$, $10^{50} + 151$. E.g. $m = 2 \cdot 3 \cdot 7 \cdot 29 \cdot 37 \cdot 211 \cdot 337 \cdot 347 \cdot 449 \cdot 467 \cdot 3527 \cdot 12613$ for $10^{50} + 151$.

Unrestricted one-shot at 256 bits: 4 seconds.

# Backup: the AKS–Bernstein certificate

- Tuple $(d, e, c, c_-, f, r, S)$: $f$ of degree $d$ over $\mathbb{Z}/p\mathbb{Z}$, $e \mid p^d - 1$; validity forces $p$ to be a prime power (Berrizbeitia, Cheng, Mihăilescu–Avanzi, Bernstein 2007).
- Every prime has one of size $n^{1+o(1)}$, found in expected $(\log p)^{2+o(1)}$, verified in $(\log p)^{4+o(1)}$. Example: $(1, 57449, 28724, 16826, y, 2, \{1\})$ for $2^{1024} + 643$.
- The asymmetry is inverted: a Las Vegas verifier gains nothing from the certificate, since it can regenerate one itself in less time than the check takes. What it buys is determinism.
- Verification is stuck at quartic: soundness forces $e = \widetilde{\Omega}(\log^2 p)$, a ring of $(\log p)^{3+o(1)}$ bits.

# Backup: descent chains with ratio $b$

- Chain of certificates with $q \le p^b$ at each step, $b \in (\frac{1}{2}, 1)$, order $c \cdot q$ with no minimality required.
- Size $O(n)/(1-b)$; verify $O(n^2 \log n)/(1-b^2)$; find $L_p(1/3, (3(1-b))^{1/3})$.
- With $b = 0.85$ on a 16-core desktop: $2^{255} - 19$ and $10^{100} + 267$ in 3.7 s each (one-shot: $\approx 15$ s and 7.9 min), and

| $p$ | find | steps | verify |
|---|---|---|---|
| 1024 bits | 128 s | 9 | 0.41 s |
| 2048 bits | 5.2 h | 12 | 2.65 s |

- $b = \frac{1}{2}$ with minimal orders is short ECPP; $b \to 1$ heads toward classical ECPP (qualitatively).